Mert Kosan

STAFF MACHINE LEARNING SCIENTIST

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Professional Experience

Visa Inc. — *Austin, TX, USA*

Staff Machine Learning Scientist — *June 2023 – Present*

- Design and develop large-scale self-supervised transformers and Large Transactional Models (LTMs) for multi-task applications spanning fraud detection, scam detection, risk modeling, and payment intelligence.
- Develop real-time machine learning models for fraud and authorized push payment scam detection using large-scale transactional, sequential, and behavioral data.
- Conduct applied research in self-supervised learning, active learning, generative AI, explainable AI, and human-in-the-loop machine learning.
- Partner with engineering and product teams to build machine learning products end-to-end, from research and prototyping through model development, system integration, validation, and production delivery.
- **Visa technologies and products:** VisaNet, Visa Direct, Visa Real-Time Payments Prevent, Large Transactional Models (LTMs), and Visa Risk Manager (VRM).

Visa Research — *Austin, TX, USA*

Ph.D. Research Intern — *Summers 2020, 2021, and 2022*

- Developed a human-in-the-loop active learning framework that models expert interests and expertise to improve data labeling, anomaly detection, and AI-assisted decision-making.
- Designed a scalable and parallelizable dynamic peer-group analysis framework compatible with existing anomaly detection methods, reducing false positives and enabling earlier detection of anomalous events.
- Developed a graph-based framework for automated fraud profiling and strategy optimization in near-real-time unsupervised fraud detection systems, generating accurate fraud profiles in under one second from limited transaction data.

Selected Research Experience

University of California, Santa Barbara — *Santa Barbara, CA, USA*

GNNX-BENCH: Benchmarking Perturbation-Based GNN Explainers — *March 2023 – June 2023*

- Led a comprehensive evaluation of seven factual and four counterfactual graph neural network explainers across multiple datasets, model architectures, and evaluation criteria.
- Identified key limitations in explanation stability, adversarial robustness, reproducibility, and predictive utility, and released an open-source benchmarking framework to support reproducible research.

University of California, Santa Barbara — *Santa Barbara, CA, USA*

GCFExplainer: Global Counterfactual Explanations for Graph Neural Networks — *October 2021 – August 2022*

- Formulated the problem of global counterfactual explanation for graph classification and developed GCFExplainer, a method that generates representative counterfactual summaries using vertex-reinforced random walks over an edit map.
- Achieved state-of-the-art performance across multiple graph classification benchmarks; the work was selected among the Top 10 papers at WSDM 2023 and received the Best Paper Award at MLoG-WSDM 2023.

Education

University of California, Santa Barbara
Santa Barbara, CA, USA

Ph.D. and M.S. in Computer Science *September 2018 – June 2023*

- Advisor: Prof. Ambuj K. Singh | GPA: 4.00/4.00.
- Dissertation: *Transparent Representation Learning for Graphs and Human-AI Collaboration.*

Sabanci University
Istanbul, Turkey

B.S. in Computer Science *September 2013 – June 2018*

- Ranked first in the program; GPA: 3.99/4.00.

Patents

Improving Machine Learning Models (2024) | **Dynamic Peer Group Analysis of Systematic Changes in Large-Scale Data** (2023) | **Auto-Profiling Anomalies** (2022)

Selected Publications

- **[KDD 2025]** *Attribute-Enhanced Similarity Ranking for Sparse Link Prediction.*
- **[ACM TIST 2024]** *GCFExplainer: Global Counterfactual Explainer for Graph Neural Networks.*
- **[KDD 2024]** *Global Human-Guided Counterfactual Explanations for Molecular Properties via Reinforcement Learning.*
- **[ICLR 2024]** *GNNX-BENCH: Unravelling the Utility of Perturbation-Based GNN Explainers through In-Depth Benchmarking.*
- **[AAAI 2024]** *DGCLUSTER: A Neural Framework for Attributed Graph Clustering via Modularity Maximization.*
- **[WSDM 2023]** *Global Counterfactual Explainer for Graph Neural Networks.* **Top 10 at WSDM; Best Paper at MLoG-WSDM**.

Academic Service

- **Leadership:** Area Chair, KDD 2025 Dataset and Benchmark Track (nine meta-reviews); Registration Chair, KDD 2023.
- **Reviewer:** More than 35 reviews for ICLR, NeurIPS, KDD, The Web Conference, ICDE, AAMAS, TNNLS, TIST, TKDD, TPAMI, and TAI.

Technical Skills

Machine Learning	Transformers, Self-Supervised Learning, Graph Neural Networks, Representation Learning, Anomaly Detection, Explainable AI, Active Learning, Generative AI
Fraud and Risk	Fraud Detection, Scam Detection, Transaction Modeling, Real-Time Payments, Behavioral Modeling, Risk Analytics
Programming	Python, PyTorch, PyTorch Geometric, Scikit-learn, SQL, C++, Java
Tools	Linux, Git, CUDA, NetworkX, Keras, LaTeX